

HASSAN MAHMOOD YOUSAFZAI

AI/ML ENGINEER

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SUMMARY

AI/ML Engineer experienced in developing multi-agent and LLM-driven systems, integrating machine learning, simulation, and autonomous reasoning for intelligent, self-governing solutions.

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript, Bash
- **AI & ML:** Scikit-learn, TensorFlow, PyTorch, LangChain, Hugging Face
- **Agentic & LLM Systems:** Multi-Agent Design, RAG, Prompt Engineering, Vector DBs (FAISS, Chroma)
- **Modeling & Simulation:** GNNs, Physics-Informed ML, Differential Equation Solvers
- **Tools & Platforms:** Git, Docker, AWS, Jupyter, VS Code

PROJECTS

- **Agentic AI Systems** – Built autonomous LLM-driven agents for coding, compliance, and research using memory, retrieval, and orchestration mechanisms in Python.
- **Material Intelligence Models** – Developed hybrid ML–simulation and GNN-based models for material property prediction and pendulum dynamics.
- **RAG for Scientific Insights** – Implemented retrieval-augmented generation pipelines for material design and knowledge discovery.
- **Sentiment Analysis** – Applied NLP and machine learning to classify sentiments from Amazon food reviews with optimized performance.

EDUCATION

Master of Science Computer Science, MS(CS)	2023
City University of Science & IT, Peshawar, PAKISTAN	
Bachelor of Science Telecommunication Engineering, BS(TE)	2011
National University of Computer & Emerging Sciences, Islamabad, PAKISTAN	

EXPERIENCE

AI & ML Engineer (Freelance)	2021 - Present
• Developed multi-agent compliance systems enabling autonomous monitoring, reasoning, and adaptive collaboration through enhanced memory and retrieval agents in Python. • Built LLM-based research and coding agents using Retrieval-Augmented Generation (RAG) for autonomous knowledge retrieval and intelligent task execution. • Created hybrid ML–simulation and GNN models for material property prediction and physics-informed system modeling. • <i>Focused on building intelligent agentic systems that combine ML, simulation, and LLM reasoning for real-world problem solving.</i>	

COURSES & CERTIFICATIONS

- Neural Networks & Deep Learning - **DeepLearning.AI** (June 2025 - Coursera)
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization–**DeepLearning.AI** (October 2025 - Coursera)